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AEROSOL GENERATING DEVICE AND HEATING STRUCTURE THEREOF

Abstract

A heating structure includes: a heating element for radiating infrared light waves in a power-on state; a tube body for the infrared light waves to pass through; and a temperature measurement unit for temperature measurement. The heating element and a tube wall of the tube body are at least partially spaced apart. The heating element includes a heating portion and a conductive portion electrically connected to the heating portion. One end of the temperature measurement unit is connected to one end of the heating portion. An other end of the temperature measurement unit is electrically connected to the conductive portion.

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION [0001] This application is a continuation of International Patent Application No. PCT/CN2023/114110, filed on Aug. 21, 2023, which claims priority to Chinese Patent Application No. 202211457367.7, filed on Nov. 17, 2022. The entire disclosure of both applications is hereby incorporated by reference herein.

FIELD

[0002] The present disclosure relates to the field of heat-not-burn atomization, and in particular, to an aerosol generating device and a heating structure thereof.

BACKGROUND

[0003] In related arts, an aerosol generating device is an electronic device used for heating an aerosol generating substrate (solid material such as tobacco and other plant leaf products) without burning it. Generally, the aerosol generating substrate is atomized within 350° C. The disadvantage of this heating method is that since the heating element conducts heat directly or indirectly through a solid material to the aerosol generating substrate, it requires that the working temperature of the heating element should not be too high, and otherwise it will cause the aerosol generating substrate to be overheated to influence the puffing taste of the aerosol generating device. Therefore, how to adapt to the working environment where the working temperature of the heating element is higher than 400° C., and how to detect and control the temperature when the heating element works under conditions higher than 400° C., are urgent problems that need to be solved by those skilled in the art.

SUMMARY

[0004] In an embodiment, the present invention provides a heating structure, comprising: a heating element configured to radiate infrared light waves in a power-on state; a tube body for the infrared light waves to pass through; and a temperature measurement unit for temperature measurement, wherein the heating element and a tube wall of the tube body are at least partially spaced apart, wherein the heating element comprises a heating portion and a conductive portion electrically connected to the heating portion, wherein one end of the temperature measurement unit is connected to one end of the heating portion, and wherein an other end of the temperature measurement unit is electrically connected to the conductive portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

[0006] FIG. 1 is a 3D schematic structural diagram of an aerosol generating device according to

some embodiments of the present disclosure.

[0007] FIG. 2 is a 3D schematic structural diagram of a heating structure of the aerosol generating device in FIG. 1.

[0008] FIG. 3 is a 3D schematic structural diagram of a temperature measurement unit of a heating element in FIG. 2.

[0009] FIG. 4 is a 3D schematic structural diagram of another temperature measurement unit of a heating element in FIG. 2.

[0010] FIG. 5 is a 3D schematic structural diagram of yet another temperature measurement unit of a heating element in FIG. 2.

[0011] FIG. 6 is a 3D schematic structural diagram of yet another temperature measurement unit of a heating element in FIG. 2.

[0012] FIG. 7 is a 3D schematic structural diagram of a heating structure according to another embodiment of the present disclosure.

[0013] FIG. 8 is a 3D schematic structural diagram of another temperature measurement unit of a heating element in FIG. 7.

[0014] FIG. 9 is a 3D schematic structural diagram of yet another temperature measurement unit of a heating element in FIG. 7.

[0015] FIG. 10 is a 3D schematic structural diagram of yet another temperature measurement unit of a heating element in FIG. 7.

DETAILED DESCRIPTION

[0016] In an embodiment, the present invention provides an improved aerosol generating device and a heating structure thereof.

[0017] In an embodiment, the present invention provides a heating structure, including a heating element configured to radiate infrared light waves in a power-on state, a tube body for the infrared light waves to pass through, and a temperature measurement unit for temperature measurement, where the heating element and the tube wall of the tube body are at least partially spaced apart, the heating element includes a heating portion and a conductive portion electrically connected to the heating portion, one end of the temperature measurement unit is connected to one end of the heating portion, and the other end of the temperature measurement unit is electrically connected to the conductive portion.

[0018] In some embodiments, the temperature measurement unit is at least partially made of a TCR material.

[0019] In some embodiments, the temperature coefficient of the TCR material is greater than 300.

[0020] In some embodiments, the heating portion includes a double-spiral section arranged longitudinally, one end of the temperature measurement unit is connected to the double-spiral section, and the other end of the temperature measurement unit is connected to the conductive portion.

[0021] In some embodiments, the temperature measurement unit is at least partially located within the double-spiral section.

[0022] In some embodiments, the temperature measurement unit is located outside the double-spiral section.

[0023] In some embodiments, the heating portion is longitudinally arranged and includes a linear first heating section and a spiral second heating section wound around the first heating section and connected to one end of the first heating section, one end of the temperature measurement unit is connected to the first heating section, and the other end of the temperature measurement unit is connected to the conductive portion.

[0024] In some embodiments, the temperature measurement unit is at least partially located within the spiral second heating section.

[0025] In some embodiments, the temperature measurement unit is located outside the spiral second heating section.

[0026] In some embodiments, the heating portion is formed by bending or winding a heating wire and includes at least one M-shaped or N-shaped section.

[0027] In some embodiments, at least part of the temperature measurement unit and the heating portion jointly form the M-shaped or N-shaped section.

[0028] In some embodiments, the heating element is arranged on the inner side of the tube body, and the heating element is spaced apart from the inner wall of the tube body.

[0029] In some embodiments, the tube body includes a first sleeve and a second sleeve sleeved over the outer periphery of the first sleeve;

[0030] a gap is provided between the first sleeve and the second sleeve, and the gap forms an accommodating cavity for accommodating the heating element; and

[0031] the heating element is arranged on the outer periphery of the first sleeve and is spaced apart from the outer wall of the first sleeve, and a heating cavity for heating an aerosol generating substrate is formed on the inner side of the first sleeve.

[0032] In some embodiments, the inner wall of the second sleeve is provided with a reflective layer for reflecting the infrared light waves.

[0033] In some embodiments, the lower part of the tube body is provided with a fixing base, and the temperature measurement unit is arranged above the fixing base or is partially overlapped with the fixing base.

[0034] In some embodiments, the highest working temperature of the heating element is 500° C.-1300° C.

[0035] In some embodiments, the present disclosure further provides an aerosol generating device, including the heating structure described above.

[0036] The present disclosure has the following beneficial effects. In the present disclosure, the heating element can radiate infrared light waves in a power-on state, and the infrared light waves can pass through the tube body to reach and heat the aerosol generating substrate. In a case that the highest working temperature of the heating element reaches 1000° C. or above (the working temperature of conventional HNB heating elements generally does not exceed 400° C.), it does not cause the aerosol generating substrate to be overheated, thereby greatly improving the puffing taste. Moreover, the preheating time is greatly reduced, thereby greatly improving the consumer experience.

[0037] The connection between the temperature measurement unit and the heating portion can quickly reflect the temperature change process of the heating portion to accurately measure the temperature of the heating element, thereby achieving the precise atomization of the aerosol generating substrate.

[0038] To provide a clearer understanding of the technical features, objectives, and effects of the present disclosure, the specific embodiments of the present disclosure will be described below with reference to the drawings.

[0039] In the description of the present disclosure, it needs to be understood that directional and positional relationships indicated by terms such as “longitudinally”, “axial”, “length”, “inner” and “outer” are directional or positional relationships based on the drawings, or directional or positional relationships formed by commonly placing the product of the present disclosure in use, which are only for the purposes of conveniently describing the present disclosure and simplifying the description, instead of indicating or implying that the device or element referred to need to have a specific orientation or be constructed and operated in a specific orientation, and cannot be understood as limitations on the present disclosure.

[0040] In addition, terms “first” and “second” are used merely for the purpose of description, and cannot be construed as indicating or implying relative importance or implying a quantity of indicated technical features. Therefore, a feature defined by “first” or “second” may explicitly indicate or implicitly include at least one of such features. In the description of the present disclosure, “multiple” means at least two, such as two or three, unless otherwise specifically

defined.

[0041] In the present disclosure, unless otherwise explicitly specified and limited, the terms “mount”, “connect”, “connection”, and “fix” are understood in a broad sense. For example, it may be a fixed connection, a detachable connection, or an integral connection; or it may be mechanical connection or electrical connection; or it may be direct connection, indirect connection through an intermediate medium, or internal communication between two elements or interaction relationship between two elements, unless otherwise specified explicitly. Those skilled in the art may understand the specific meanings of the terms in the present disclosure based on specific situations.

[0042] FIG. 1 shows an aerosol generating device **1** and an aerosol generating substrate **2** detachably inserted into one end of the aerosol generating device **1** in some embodiments of the present disclosure. In some embodiments, the aerosol generating device **1** may be in a square column shape for ease of gripping by the user, and may be configured to perform low-temperature baking and heating on the aerosol generating substrate **2** inserted therein to release an aerosol extract from the aerosol generating substrate **2** in a non-burning state. In addition, the atomization stability is good and the puffing taste is excellent. In some embodiments, the aerosol generating substrate **2** may be cylindrical and may be a filamentous or sheet-like solid material made from leaves and/or stems of plants, and aroma components may be further added to the solid material. As can be understood, the aerosol generating device **1** is not limited to be in a square column shape. In some other embodiments, it may also in any other shape such as cylindrical shape or elliptical column shape.

[0043] In some embodiments, the aerosol generating device **1** may include a heating structure **10** and a housing **20** for carrying the heating structure **10**. In some embodiments, the heating structure **10** may be cylindrical and may allow the aerosol generating substrate **2** to be detachably inserted to heat and bake the aerosol generating substrate **2** from the outer periphery. In some embodiments, the aerosol generating device **1** may further include a power supply assembly arranged in the housing **20**. The heating structure **10** may be partially inserted into the aerosol generating substrate **2**. Specifically, it may be partially inserted into the substrate section of the aerosol generating substrate **2**, and generates thermal radiation to heat the substrate section of the aerosol generating substrate **2** in a power-on state, causing it to be atomized to generate an aerosol. In this embodiment, the thermal radiation may be infrared radiation. The heating structure **10** has the advantages of easiness in assembling, simple structure, high atomization efficiency, strong stability, and long service life. The power supply assembly is electrically connected to the heating structure **10** to supply power to the heating structure **10**.

[0044] Referring to FIG. 2 together, in some embodiments, the heating structure **10** may include a tube body **11** for infrared light waves to pass through, a heating element **12** configured to radiate infrared light waves in a power-on state, a temperature measurement unit **13**, and a fixing base **14**. The tube body **11** covers at least part of the heating element **12** and may allow light waves to pass through to reach the aerosol generating substrate **2**. Specifically, in this embodiment, the tube body **11** may allow infrared light waves to pass through, thereby facilitating the heating element **12** to radiate infrared light waves to heat the aerosol generating substrate **2**. Specifically, a gap is provided between the heating element **12** and the tube body **11**. In a power-on state, the heating element **12** quickly heats to increase the temperature up to 1000-1300° C. within 1-3 s, the surface temperature of the tube body **11** can be controlled below 350° C., and the overall atomization temperature of the aerosol generating substrate **2** is controlled at 300-350° C., thereby achieving the precise atomization of the aerosol generating substrate **2** mainly in a wave band of 2-5 μm . In some embodiments, the heating element **12** may include a heating portion **121** configured to radiate infrared light waves in a power-on state, and a conductive portion **122** arranged at one end of the heating portion **121** and configured to connect to power. One end of the temperature measurement unit **13** is connected to one end of the heating portion **121**, and the conductive portion **122** is connected to the other end of the temperature measurement unit **13** and/or the heating portion **121**.

The connection between the temperature measurement unit **13** and the heating portion **121** can quickly reflect the temperature change process of the heating portion **121** of the heating element **12** to accurately measure the temperature of the heating element **12**, thereby achieving the precise atomization of the aerosol generating substrate **2**. The fixing base **14** is arranged at a lower open end of the tube body **11**, and the temperature measurement unit **13** is arranged above the fixing base **14** or is partially overlapped with the fixing base **14**. The conductive portion **122** of the heating element **12** penetrates through the fixing base **14** to connect to power. The highest working temperature of the heating element **12** is 500° C.-1300° C., which is much higher than 400° C. in the existing technology, thereby solving the problems of easy burning and inconsistent taste in high-temperature working environments, and greatly shortening the preheating time.

[0045] In some embodiments, the tube body **11** may be a quartz glass tube. Of course, as can be understood, in some other embodiments, the tube body **11** is not limited to a quartz tube and may be any other window material that allows light waves to pass through, such as infrared transparent glass, transparent ceramics, or diamond.

[0046] In some embodiments, the tube body **11** may be a hollow tube. Specifically, the tube body **11** includes a tubular body **111** with a circular cross section, and a pointed structure **112** arranged at one end of the tubular body **111**. Of course, as can be understood, in some other embodiments, the cross section of the tubular body **111** is not limited to be circular. The tubular body **111** is a hollow structure with an opening in one end. The tube body **11** may be mounted on the fixing base **14**. Specifically, the tube body **11** may be partially inserted into the fixing base **14**. The opening of the tube body may be located in the fixing base **14**. The pointed structure **112** is arranged at the end of the tubular body **111** away from the opening. By arranging the pointed structure **112**, it is convenient for at least part of the heating structure **10** to be inserted into the aerosol generating substrate **2**. In this embodiment, a first accommodating cavity **113** is formed on the inner side of the tube body **11**. The first accommodating cavity **113** is a cylindrical cavity and may be non-sealed. When the heating element **12** is mounted in it, the first accommodating cavity **113** does not need to be vacuumed or filled with inert gas. Of course, in order to further improve the taste and the service life of the heating element **12**, the tube body **11** may also be sealed or vacuumed. As can be understood, in some other embodiments, the heating element **12** may also be arranged on the outer periphery of the tube body **11** in a manner of being spaced apart, and a second accommodating cavity for accommodating the aerosol generating substrate **2** may be formed on the inner side of the tube body **11**. In this embodiment, the tube body **11** further includes a positioning portion. The positioning portion is arranged at the opening of the tubular body **111** and may extend radially outwards along the tubular body **111** to form a positioning flange for the mounting and positioning of the tube body **11** and the fixing base **14**. In this embodiment, the positioning portion may be integrally formed with the tubular body **111**. Of course, as can be understood, in some other embodiments, the positioning portion may be detachably assembled with the tube body **11** by sleeving, screwing, or clamping, for example. In this embodiment, a gap is provided between the inner wall of the tube body **11** and the heating element **12**. The gap may be filled with air or maintained in a vacuum state. By providing the gap, there can be no direct contact between the tube body **11** and the heating element **12**.

[0047] In some embodiments, the number of the heating element **12** may be one, which may be longitudinally arranged and wound to form a heating portion **121** that is spiral on the whole. Specifically, the heating element **12** may be cylindrical on the whole and may be wound to form a single-spiral structure, double-spiral structure, M-shaped structure, N-shaped structure, or structure with any other shape. Of course, as can be understood, in some other embodiments, the number of the heating element **12** is not limited to one, and may be two or more. The shape of the heating element **12** is not limited to a cylindrical shape. In some embodiments, the shape of the heating element **12** may be a sheet shape.

[0048] In this embodiment, the heating portion **121** may be placed in the tube body **11**, spaced apart

from the tube wall of the tube body **11** on the whole, and configured to radiate infrared light waves in a power-on state. The infrared light waves may pass through the tube body **11** to reach the aerosol generating substrate **2**. Of course, as can be understood, in some other embodiments, the heating portion **121** may also be partially spaced apart from the tube wall of the tube body **11**. In this embodiment, the heating portion **121** may be in a longitudinally elongated spiral shape. Of course, as can be understood, in some other embodiments, the heating portion **121** is not limited to a spiral shape.

[0049] In this embodiment, one end of the heating portion **121** is provided with a conductive portion **122**. The conductive portion **122** is connected to the heating portion **121**, may be led out from the opening of the tube body **11**, penetrate through the fixing base **14**, and be electrically connected to a power supply assembly. In this embodiment, the conductive portion **122** may be fixed to the heating portion **121** through welding to form an integrated structure. Of course, as can be understood, in some other embodiments, the heating portion **121** may be integrally formed with the conductive portion **122**. In this embodiment, the number of the conductive portions **122** may be two. The two conductive portions **122** may be spaced apart, be respectively connected to two ends of the heating portion **121**, extend towards the same end, and penetrate through the tube body **11** from the opening of one end of the tube body **11**. In this embodiment, the conductive portion **122** may be a lead wire, which may be welded to the heating portion **121**. Of course, as can be understood, in some other embodiments, the conductive portion **122** is not limited to a lead wire and may be any other conductive structure. By arranging the conductive portion **122** at one end of the heating portion **121** and leading it out from the tube body **11**, the entire heating structure **10** can be conveniently assembled, thereby simplifying the assembling process. During assembling, the heating structure **10** may be mounted on a supporting base, and then enabled to be in contact with a conductive member located in the supporting base.

[0050] In some embodiments, the heating element **12** may include a heating substrate for generating heat in a power-on state, and an infrared radiation layer. The heating substrate may generate heat in a power-on state. The infrared radiation layer is arranged on the outer surface of the heating substrate and is configured to radiate the heat generated by the heating substrate. In this embodiment, the heating substrate and infrared radiation layer are concentrically distributed on the cross section of the heating portion.

[0051] In this embodiment, the heating substrate may be cylindrical on the whole. Specifically, the heating substrate may be a heating wire. Of course, as can be understood, in some other embodiments, the heating substrate may not be limited to be cylindrical, but may be in a sheet shape, that is, the heating substrate may be a heating sheet. The heating substrate includes a metal substrate with high-temperature oxidation resistance. The metal substrate may be a metal wire. Specifically, the heating substrate may be a nickel-chromium alloy substrate (such as nickel-chromium alloy wire), iron-chromium-aluminum alloy substrate (such as iron-chromium-aluminum alloy wire) or the like made of a metal material with good high-temperature oxidation resistance, high stability, and good deformation resistance. In this embodiment, the radial size of the heating substrate may be 0.15 mm-0.8 mm.

[0052] In this embodiment, the heating element **12** further includes an antioxidant layer, and the antioxidant layer is formed between the heating substrate and the infrared radiation layer. Specifically, the antioxidant layer may be an oxide film, the heating substrate undergoes high-temperature heat treatment to form a dense oxide film on its own surface, and the oxide film forms the antioxidant layer. Of course, as can be understood, in some other embodiments, the antioxidant layer is not limited to including an oxide film formed by itself. In some other embodiments, it may be an antioxidant coating layer coated on the outer surface of the heating substrate. By forming the antioxidant layer, it can ensure that the heating substrate is not or is rarely oxidized when heated in the air environment, thereby improving the stability of the heating substrate. Therefore, there is no need to vacuum and fill inert gas or reducing gas into the first accommodating cavity **113**, nor to

seal the opening, thereby simplifying the assembling process of the entire heating structure **10** and reducing the manufacturing cost. In this embodiment, the thickness of the antioxidant layer may be selectively 1 μm -150 μm . When the thickness of the antioxidant layer is less than 1 μm , the heating substrate is easily oxidized. When the thickness of the antioxidant layer is greater than 150 μm , it will reduce the heat conduction between the heating substrate and the infrared radiation layer.

[0053] In this embodiment, the infrared radiation layer may be an infrared layer. The infrared layer may be an infrared layer-forming substrate formed on the side of the antioxidant layer away from the heating substrate under high-temperature heat treatment. In this embodiment, the infrared layer-forming substrate may be a silicon carbide, spinel, or composite substrate. Of course, as can be understood, in some other embodiments, the infrared radiation layer is not limited to an infrared layer. In some other embodiments, the infrared radiation layer may be a composite infrared layer. In this embodiment, the infrared layer may be formed on the side of the antioxidant layer away from the heating substrate through dip coating, spray coating, brush coating, and other methods. The thickness of the infrared radiation layer may be 10 μm -300 μm . When the thickness of the infrared radiation layer is between 10 μm -300 μm , the thermal radiation effect is better, and the atomization efficiency and taste of the aerosol generating substrate **2** are better. Of course, as can be understood, in some other embodiments, the thickness of the infrared radiation layer is not limited to 10 μm -300 μm .

[0054] In some embodiments, the heating element **12** further includes a bonding layer arranged between the antioxidant layer and the infrared radiation layer. The bonding layer may be configured to prevent the heating substrate from being broken down locally, thereby further improving the bonding strength between the antioxidant layer and the infrared radiation layer. In some embodiments, the bond in the bonding layer may be glass powder, that is, the bonding layer may be a glass powder layer.

[0055] Referring to FIG. 3 to FIG. 6 together, in some embodiments, the heating portion **121** may be longitudinally arranged, and include a linear first heating section **1211** and/or a spiral second heating section **1212** wound around the first heating section **1211** and connected to the first heating section **1211**. In some embodiments, the conductive portion **122** may include a first lead wire section **1221** connected to the first heating section **1211** and/or a second lead wire section **1222** connected to the second heating section **1212**. In some embodiments, the temperature measurement unit **13** may at least partially replace the first heating section **1211** and/or the second heating section **1212** and/or the first lead wire section **1221** and/or the second lead wire section **1222**, the temperature measurement unit **13** is connected to the heating portion **121**, and the first heating section **1211**, the second heating section **1212**, the first lead wire section **1221**, and the second lead wire section **1222** may be integrally formed or welded together. The temperature measurement unit **13** can quickly reflect the temperature change process of the heating portion **121** of the heating element **12** to accurately measure the temperature of the heating portion **121**, so as to control the overall atomization temperature of the aerosol generating substrate **2** at 300-350° C., thereby achieving the precise atomization of the aerosol generating substrate **2** mainly in a wave band of 2-5 μm . Specifically, referring to FIG. 3, in some embodiments, the heating portion **121** includes a spiral second heating section **1212**. In some embodiments, the conductive portion **122** may include a first lead wire section **1221** and a second lead wire section **1222**, and the temperature measurement unit **13** may replace the first heating section **1211** connected to the second heating section **1212**. Referring to FIG. 4, the heating portion **121** includes a spiral second heating section **1212**, the conductive portion **122** includes a second lead wire section **1222**, and the temperature measurement unit **13** may replace the first heating section **1211** and the first lead wire section **1221**. Referring to FIG. 5, the heating portion **121** includes a linear first heating section **1211** and a spiral second heating section **1212** wound around the first heating section **1211** and connected to the first heating section **1211**, the conductive portion **122** includes a first lead wire section **1221**, and the temperature measurement unit **13** may completely replace the second lead wire section **1222**.

Referring to FIG. 6, in some embodiments, the heating portion **121** includes a linear first heating section **1211** and a spiral second heating section **1212** wound around the first heating section **1211** and connected to the first heating section **1211**, the temperature measurement unit **13** partially replaces the second lead wire section **1222**, this part of the second lead wire section **1222** is connected to the second heating section **1212**, and the conductive portion **122** includes a first lead wire section **1221** and another part of the second lead wire section **1222**. As can be understood, a connection position between the temperature measurement unit **13** and the heating portion **121** may also be set as needed.

[0056] In some other embodiments, the heating portion **121** may further include a double-spiral section arranged longitudinally, one end of the temperature measurement unit **13** is connected to the double-spiral section, and the other end of the temperature measurement unit **13** is connected to the conductive portion **122**. Specifically, the conductive portion **122** includes a third lead wire section and a fourth lead wire section respectively connected to the double-spiral section. Specifically, the temperature measurement unit **13** may at least partially replace the third heating section. Alternatively, the temperature measurement unit **13** may also at least partially replace the third lead wire section and/or fourth lead wire section connected to the third heating section. As can be understood, a connection position between the temperature measurement unit **13** and the heating portion **121** may also be set as needed.

[0057] In yet another embodiment, the heating portion **121** may include multiple M-shaped or N-shaped sections formed by winding or bending strip-shaped or linear heating wires, and the conductive portion includes a fifth lead wire section and/or a sixth lead wire section respectively connected to the M-shaped or N-shaped sections. In some embodiments, the temperature measurement unit **13** may at least partially replace the M-shaped or N-shaped sections. Alternatively, the temperature measurement unit **13** may also replace the fifth lead wire section and/or sixth lead wire section connected to the M-shaped or N-shaped sections. As can be understood, a connection position between the temperature measurement unit **13** and the heating portion **121** may also be set as needed.

[0058] In some embodiments, the temperature measurement unit **13** is at least partially made of a TCR (temperature coefficient of resistance) material. In some embodiments, the temperature coefficient of the TCR material is greater than 300, thereby making the measurement data more accurate. As can be understood, the temperature coefficient of the TCR material may also be set to be less than 300. In some embodiments, the TCR material may be strip-shaped. As can be understood, the TCR material is not limited to be strip-shaped, and it may also be sheetlike, columnar, spiral, etc.

[0059] In some embodiments, the fixing base **14** may be made of ceramic or PEEK high-temperature insulating material. It may include two fixing through holes **141** provided in the fixing base **14**. The two fixing through holes **141** are configured to allow two conductive portions **122** to be inserted in.

[0060] In some embodiments, the heating structure further includes a supporting rod. The supporting rod is an insulating rod. The supporting rod may partially penetrate into the heating portion **121**, may be located at the center of the heating portion **121**, and may be insulated from the heating portion **121**, and can play a role of supporting the heating portion **121**. By arranging the supporting rod, the heating portion **121** can be supported to ensure that the heating element **12** is not completely deformed due to heating, thereby ensuring that the gap between the heating element **12** and the tube body **11** is uniform, and further ensuring the consistency of the temperature field. As can be understood, in some other embodiments, the supporting rod may not be arranged, but other structures may be arranged to support the heating portion **121**.

[0061] FIG. 7 shows a heating structure **10a** in a second embodiment of the present disclosure, whose difference from that in the first embodiment lies in that the heating structure **10a** is not limited to being partially inserted into the aerosol generating substrate to heat the aerosol

generating substrate. In this embodiment, the heating structure **10a** may be sleeved over the outer periphery of the substrate section of the aerosol generating substrate, and adopt a circumferential heating method to heat the aerosol generating substrate. In some embodiments, the heating structure **10a** may include a tube body **11a**, a heating element **12a**, and a temperature measurement unit **13a**. The heating element **12a** and the tube wall of the tube body **11a** are at least partially spaced apart. In some embodiments, the heating element **12a** may include a heating portion **121a** configured to radiate infrared light waves in a power-on state, and a conductive portion **122a** arranged at one end of the heating portion **121a** and configured to connect to power.

[0062] In some embodiments, the tube body **11a** may include a first sleeve **111a** and a second sleeve **112a** sleeved over the outer periphery of the first sleeve **111a**. The first sleeve **111a** is a hollow structure with two ends that run through. The first sleeve **111a** may be cylindrical, and the inner diameter thereof is slightly larger than the outer diameter of the aerosol generating substrate. A gap is provided between the first sleeve **111a** and the second sleeve **112a**, and the gap forms an accommodating cavity for accommodating the heating element **12a**. The axial length of the first sleeve **111a** may be greater than the axial length of the second sleeve **112a**. The second sleeve **112a** may be sleeved over the outer periphery of the first sleeve **111a**. The second sleeve **112a** may be cylindrical. The radial size of the second sleeve **112a** may be larger than the radial size of the first sleeve **111a**. In some embodiments, the heating element **12a** is wound around the outer periphery of the first sleeve **111a** and spaced apart from the outer wall of the second sleeve **112a**, thereby creating a certain temperature difference between the inner wall of the accommodating cavity and the heating element **12** to achieve a thermal insulation effect. A heating cavity for heating the aerosol generating substrate is formed on the inner side of the first sleeve **111a**.

[0063] In some embodiments, the inner side of the second sleeve **112a** may be provided with a reflective layer, and the reflective layer is configured to reflect the heat of the heating element **12** and radiate it to the aerosol generating substrate to enhance the heating energy efficiency. As can be understood, the first sleeve **111a** and the second sleeve **112a** are not limited to be cylindrical, and they may also be in any other shape such as square column shape or elliptical column shape.

[0064] In some embodiments, the second sleeve **112a** may further include a fixing structure, and the fixing structure is configured to fix the heating element **12a**.

[0065] Referring to FIG. 8 to FIG. 10 together, in some embodiments, the heating portion **121a** may be a spiral fourth heating section **1211a**, and the conductive portion **122a** includes a seventh lead wire section **1221a** and/or an eighth lead wire section **1222a** respectively connected to two opposite sides of the fourth heating section **1211a**. The temperature measurement unit **13a** may at least partially replace the seventh lead wire section **1221a** and/or the eighth lead wire section **1222a** connected to the heating portion **121a**. The temperature measurement unit **13a** can quickly reflect the temperature change process of the heating portion **121a** of the heating element **12a** to accurately measure the temperature of the heating element **12a**, thereby achieving the precise atomization of the aerosol generating substrate. Specifically, referring to FIG. 8, in some embodiments, the temperature measurement unit **13a** may completely replace the seventh lead wire section **1221a**, and the conductive portion **122a** includes the eighth lead wire section **1222a**. Referring to FIG. 9, in some embodiments, the temperature measurement unit **13a** may partially replace the seventh lead wire section **1221a** connected to the heating portion **121a**, and the conductive portion **122a** includes the eighth lead wire section **1222a**. Referring to FIG. 10, the temperature measurement unit **13a** may partially replace the seventh lead wire section **1221a** connected to the heating portion **121a** and completely replace the eighth lead wire section **1222a**. The conductive portion **122a** includes another part of the seventh lead wire section **1221a**. As can be understood, a connection position between the temperature measurement unit **13a** and the heating portion **121a** may also be set as needed.

[0066] While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or

exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

[0067] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. A heating structure, comprising: a heating element configured to radiate infrared light waves in a power-on state; a tube body for the infrared light waves to pass through; and a temperature measurement unit for temperature measurement, wherein the heating element and a tube wall of the tube body are at least partially spaced apart, wherein the heating element comprises a heating portion and a conductive portion electrically connected to the heating portion, wherein one end of the temperature measurement unit is connected to one end of the heating portion, and wherein an other end of the temperature measurement unit is electrically connected to the conductive portion.
2. The heating structure of claim 1, wherein the temperature measurement unit at least partially comprises a TCR material.
3. The heating structure of claim 2, wherein a temperature coefficient of the TCR material is greater than 300.
4. The heating structure of claim 2, wherein the heating portion comprises a double-spiral section arranged longitudinally, wherein one end of the temperature measurement unit is connected to the double-spiral section, and wherein an other end of the temperature measurement unit is connected to the conductive portion.
5. The heating structure of claim 4, wherein the temperature measurement unit is at least partially located within the double-spiral section.
6. The heating structure of claim 4, wherein the temperature measurement unit is located outside the double-spiral section.
7. The heating structure of claim 2, wherein the heating portion is longitudinally arranged and comprises a linear first heating section and a spiral second heating section wound around the first heating section and connected to one end of the first heating section, wherein one end of the temperature measurement unit is connected to the first heating section, and wherein an other end of the temperature measurement unit is connected to the conductive portion.
8. The heating structure of claim 7, wherein the temperature measurement unit is at least partially located within the spiral second heating section.
9. The heating structure of claim 7, wherein the temperature measurement unit is located outside the spiral second heating section.
10. The heating structure of claim 2, wherein the heating portion is formed by bending or winding a heating wire and comprises at least one M-shaped or N-shaped section.

- 11.** The heating structure of claim 10, wherein at least part of the temperature measurement unit and the heating portion jointly form the M-shaped or N-shaped section.
- 12.** The heating structure of claim 1, wherein the heating element is arranged on an inner side of the tube body, and wherein the heating element is spaced apart from an inner wall of the tube body.
- 13.** The heating structure of claim 1, wherein the tube body comprises a first sleeve and a second sleeve sleeved over an outer periphery of the first sleeve, wherein a gap is provided between the first sleeve and the second sleeve, and the gap forms an accommodating cavity configured to accommodate the heating element, wherein the heating element is arranged on an outer periphery of the first sleeve and is spaced apart from an outer wall of the first sleeve, and wherein a heating cavity configured to heat an aerosol generating substrate is formed on an inner side of the first sleeve.
- 14.** The heating structure of claim 13, wherein an inner wall of the second sleeve is provided with a reflective layer configured to reflect the infrared light waves.
- 15.** The heating structure of claim 1, wherein a lower part of the tube body is provided with a fixing base, and wherein the temperature measurement unit is arranged above the fixing base or is partially overlapped with the fixing base.
- 16.** The heating structure of claim 1, wherein a highest working temperature of the heating element is 500° C.-1300° C.
- 17.** An aerosol generating device, comprising: the heating structure of claim 1.
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